About 2,600 results for “review wildfire spread prediction”

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Efficient Cloud-Based Calibration of Input Data for Forest Fire Spread Prediction

Edigley Fraga | A. Cortés | T. Margalef | P. Hernández | Environmental Science, Computer Science · IEEE International Conference on e-Science · 1 October 2022

TLDR A new fitness function in tandem with a strict deadline policy that decreases overall processing time is proposed, improving the genetic algorithm convergence and decreasing the response time for the calibration stage, setting up an important upper bound limit to the critical compute-intensive adjustment phase. Expand

1 | PDF | IEEE | Save | Cite

Prediction of Wildfire Spread and Propagation Algorithm for Disaster Area

Nam-kyoung Koo | Kang-whan Lee | Environmental Science · 2016

TLDR The forest fire diffusion model and **predictions** of the path identified to ensure the safe zone and algorithm could provide the technique to analyze the attribute information that temperature, wind, smoke measured over time. Expand

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Dynamic correction of forest fire spread prediction using observation error covariance matrix estimation technique based on FLC-GRU

Tianyu Wu | Qixing Zhang | Jiping Zhu | Jinhong Wu | Jinyang Dai | Yongming Zhang

Environmental Science, Engineering · Fire Ecology · 14 October 2024

Data assimilation (DA) techniques have played a significant role in improving the **prediction** accuracy of forest fire **spread**. The dynamic correction technique weights the predicted and observed… Expand

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Exploration of geo-spatial data and machine learning algorithms for robust wildfire occurrence prediction

S. Illarionova | Dmitrii G. Shadrin | +5 authors | Evgeny Burnaev

Environmental Science, Computer Science · Scientific Reports · 28 March 2025

TLDR A unified pipeline for data acquisition and subsequent ML-based algorithm development is developed, incorporating the following algorithms: Random Forest, XGBoost, Autoencoder, ConvLSTM, Attention Multilayer Perceptron, and RegNetX, which explores the potential of integrating such algorithms into emergency response systems. Expand

2 | PubMed | Save | Cite

Forest fire spread prediction using deep learning

Fadoua Khennou | Jade Ghaoui | M. Akhloufi | Environmental Science, Computer Science · Defense + Commercial Sensing · 12 April 2021

TLDR FU-NetCast, a deep learning model based on U-Net, past **wildfires** events and weather data, which uses an intelligent model to study forest fire **spread** over a period of 24 hours and achieves an accuracy of 92.73% and an AUC of 80% Expand

22 | SPIE | Save | Cite

Generative Algorithms for Fusion of Physics-Based Wildfire Spread Models with Satellite Data for Initializing Wildfire Forecasts

B. Shaddy | Deep Ray | +7 authors | A. Oberai | Environmental Science, Physics · Artificial Intelligence for the Earth Systems · 5 September 2023

TLDR This work develops a physics-informed approach for inferring the history of a **wildfire** from satellite measurements, providing the necessary information to initialize coupled atmosphere-**wildfire** models from a measured **wildfire** state. Expand

8 [PDF] arXiv Save Cite

Uncertainty quantification of forecast error in coupled fire–atmosphere wildfire spread simulations: sensitivity to the spatial resolution

U. Ciri M. M. Garimella F. Bernardoni R. L. Bennett S. Leonardi Environmental Science · International journal of wildland fire · 18 August 2021

A methodology to quantify uncertainty in **wildfire** forecast using coupled fire-atmosphere computational models is presented. In these models, an atmospheric solver is coupled with a fire-**spread**... Expand

4 Publisher Save Cite

Cloud-Based Urgent Computing for Forest Fire Spread Prediction under Data Uncertainties

Edigley Fraga A. Cortés T. Margalef P. Hernández Environmental Science, Computer Science · International Conference on High Performance... · 1 December 2021

TLDR This work proposes a novel forest fire **spread prediction** platform based on a proven two-stage **prediction** model devised to deal with input data uncertainties, able to calibrate the unknown parameters based on the real observed data using an iterative process. Expand

2 IEEE Save Cite

Assessment and Validation of a Computer Vision Algorithm for Wildfire Rate of Spread Estimation

Ehsan Ameri Kyle Awayan Jeanette Cobian-Iñiguez Environmental Science, Computer Science · Fire · 5 December 2024

TLDR A computer vision algorithm is developed and validated for fire **spread** estimation in pristine visual environments with varying accuracy depending on the fuel type and results show that the algorithm estimates fire **spread** well in pristine visual environments with varying accuracy depending on the fuel type. Expand

Publisher Save Cite

Towards a Comprehensive Dataset for Next-Day Wildfire Prediction

Syed Haider Ali Anish Goel Aditya Singirikonda Ali Y Khan Ting Xiao

Environmental Science, Computer Science · IEEE International Conference on Software Quality... · 1 December 2022

TLDR A large-scale multivariate dataset is produced to develop deep learning models to understand and forecast the **spread** of **wildfire**s and it is expected that this approach can be utilized to produce state of the artDeep learning models for other natural catastrophes as well. Expand

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